

GRANT COUNTY AP, NM, USA

WMO: 722721

Lat: 32.633N

Lon: 108.167W

Elev: 5373

StdP: 12.06

Time Zone: -7.00 (NAM)

Period: 96-19

WBAN: 93063

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	18.4	22.6	-4.6	5.2	43.5	-0.4	6.5	48.2	28.5	44.1	26.0	46.8	6.5	340	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	31.6	98.5	61.0	94.8	60.1	91.6	59.7	68.4	80.7	67.3	79.3	66.1	78.4	10.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.5	115.4	72.1	63.8	108.4	70.9	62.9	105.2	70.9	36.0	81.2	34.9	79.1	33.9	77.7	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.2	22.9	19.4	DB	11.3	100.8	5.9	3.3	7.0	103.2	3.6	105.1	0.3	107.0	-4.0	109.4	
			WB	8.8	69.3	5.7	2.5	4.7	71.1	1.3	72.5	-1.9	73.9	-6.0	75.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.6	42.2	45.7	52.2	58.2	65.8	76.3	77.8	75.2	69.9	60.1	49.8	41.6
	DBStd	14.28	6.82	7.74	6.40	6.37	7.04	4.86	4.48	3.82	4.56	7.06	7.41	7.24
	HDD50	843	251	155	51	12	1	0	0	0	0	10	93	270
	HDD65	3309	707	541	398	213	78	0	0	0	11	182	455	724
	CDD50	4363	9	34	119	259	490	790	862	781	597	323	88	11
	CDD65	1352	0	0	0	9	102	340	397	316	158	30	0	0
	CDH74	17609	0	19	108	532	2072	5017	4262	3109	1817	576	96	1
	CDH80	8105	0	2	12	113	860	2761	2168	1368	661	151	9	0
(14) Wind	WSAvg	8.9	8.2	9.3	10.1	10.9	10.1	9.8	8.7	7.7	7.7	8.1	8.0	8.0
(15) Precipitation	PrecAvg	15.9	1.2	1.1	0.6	0.4	0.5	0.7	3.1	3.1	2.0	1.1	0.8	1.2
	PrecMax	20.6	4.9	4.1	1.9	2.7	3.6	3.4	5.9	5.8	3.2	4.0	3.9	4.3
	PrecMin	9.6	0.2	0.0	0.1	0.1	0.0	0.0	1.0	1.8	0.1	0.0	0.0	0.0
	PrecStd	2.8	1.2	0.9	0.5	0.6	0.8	0.7	1.4	1.1	0.9	0.9	0.9	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.5	76.6	81.5	87.8	97.0	100.8	100.4	96.8	92.9	89.8	81.4	71.7
		MCWB	47.4	49.1	49.8	52.0	57.7	60.3	61.9	63.5	60.1	59.4	52.6	49.2
	2%	DB	65.7	71.8	77.0	82.5	91.2	98.9	97.4	92.9	90.2	84.0	76.8	65.6
		MCWB	44.9	47.1	48.4	50.8	55.8	59.7	63.3	62.4	60.4	58.2	50.7	45.4
	5%	DB	62.2	66.5	73.1	80.6	88.5	95.4	93.4	90.4	87.0	80.9	72.3	62.6
		MCWB	43.4	45.0	47.2	50.0	54.3	58.3	62.3	62.7	60.2	55.7	49.3	43.9
	10%	DB	57.3	63.3	70.3	76.8	84.3	92.7	90.8	87.9	83.7	76.7	67.5	57.1
		MCWB	41.4	43.6	45.9	48.5	53.0	57.7	62.1	63.0	59.3	53.2	47.7	42.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.6	51.4	52.3	55.4	60.3	66.0	70.5	70.0	67.4	64.4	54.9	52.2
		MCDB	56.6	67.7	72.6	76.0	83.5	80.8	82.7	82.2	80.0	78.6	72.4	64.3
	2%	WB	48.4	49.6	50.1	53.0	58.4	64.0	68.9	68.3	65.6	61.5	52.5	48.1
		MCDB	59.0	64.6	71.2	76.7	82.7	83.8	81.5	80.3	76.9	75.5	71.9	60.5
	5%	WB	45.7	47.3	48.5	51.1	56.3	62.1	67.7	67.2	64.3	59.2	50.7	45.6
		MCDB	57.0	60.6	69.3	76.1	82.5	85.1	79.7	79.1	75.6	72.9	68.9	59.3
	10%	WB	43.4	45.2	46.9	49.6	54.4	60.6	66.4	66.1	63.0	57.0	48.7	43.2
		MCDB	55.6	59.7	67.3	74.2	81.7	85.0	79.1	77.8	74.7	71.6	64.9	55.5
(35) Mean Daily Temperature Range	5% DB	MDBR	28.0	29.1	31.2	32.4	33.0	31.6	24.9	24.2	25.9	29.6	30.3	27.4
		MCDBR	35.0	36.1	36.2	37.2	36.6	33.3	28.8	28.0	30.7	33.1	37.6	33.8
	5% WB	MCWBR	19.2	17.8	16.9	17.1	15.5	12.9	8.3	8.0	10.5	14.4	18.8	18.5
		MCDWR	28.8	29.1	32.4	33.7	33.5	29.2	23.5	22.4	22.7	27.1	33.5	29.1
(40) Clear-Sky Solar Irradiance	taub	taub	0.237	0.241	0.259	0.280	0.296	0.322	0.366	0.344	0.311	0.269	0.246	0.237
		taud	2.693	2.694	2.598	2.513	2.515	2.439	2.424	2.457	2.549	2.649	2.678	2.686
	Ebn at Noon	Ebn at Noon	322	329	327	320	313	304	290	296	304	313	316	317
		Edh at Noon	22	24	29	33	34	36	37	35	30	25	22	21
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1124	1442	1899	2326	2557	2537	2199	2066	1859	1567	1239	1032
	RadStd	RadStd	92	125	110	91	100	142	104	65	93	120	68	52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	-135	+162	N/A

Nomenclature: See separate page